

AMGPSU-I12-P24A

INDUSTRIAL DIN-RAIL

24W POWER SUPPLY



Industrial Power Solutions

AMG's industrial DIN-Rail 24W power supplies provide reliable power for AMG standard products and ensure stable equipment operation over a wide temperature range. They are suitable for all AMG standard products (depending on voltage).



			
PSU 24W 12V (11-14)	Temp -25~+70°C	Mounting DIN	NDAA Compliant

[AMGPSU-I12-P24A]

OVERVIEW

Designed in a compact, robust DIN rail housing, the AMGPSU-I12-P24 series industrial power supplies are ideally suited for powering AMG standard Ethernet and analogue fiber transmission equipment. Its wide operating temperature range ensures reliable operation in even the harshest environments.

With a native 12V output and a user adjustable range of 11-14V ensure the correct power output voltage is available for most standard AMG equipment.

The power supply offers a high level of stability and immunity to noise and a low ripple for best in class performance.

Compliant to the international EN55032 Class B standards for EMC and are safety approved to UL62368-1 and BS EN / EN62368-1.

A wide voltage input range that features dual-use inputs for both DC and AC voltages that support 90-264V_{AC} or 120-370V_{DC} ensures the widest possible site support.

A range of other output power levels are available within the AMGPSU product range.

FEATURES

- Compact size – ideal for confined spaces, including camera poles and roadside cabinets
- -25°C to +70°C temperature maintains performance in extreme conditions
- DIN rail mountable – quick to install and remove for maintenance
- High efficiency - up to 90% typical
- Universal 90-264V_{AC} or 120-370V_{DC} input range
- Output short circuit, over-current and over-voltage protection included as standard
- High I/O isolation test voltage up to 4000V_{AC}
- Low ripple & noise
- Withstand 300V_{AC} surge input for 5 seconds
- BS EN / EN62368 & UL safety approved
- AMG 3 Year Support Warranty

Specifications.

Input.

Characteristics	Operating Conditions	Min.	Typ.	Max.	Unit
Input Voltage Range	AC Input	90	-	264	VAC
	DC Input	120	-	370	VDC
Input Frequency		47	-	63	Hz
Input Current	115VAC	-	-	0.88	A
	230VAC	-	-	0.48	
Inrush Current	115VAC	-	25	-	
	230VAC	-	45	-	
Leakage Current	264VAC	0.25mA RMS max.			
Connector		2-Way Screw Terminal			

Output.

Characteristics	Operating Conditions		Min.	Typ.	Max.	Unit
Output Voltage Accuracy	0% - 100% Load		-	±2	-	%
Line Regulation	Rated Load		-	±1	-	
Load Regulation	230VAC		-	±1	-	
Output Ripple & Noise	20MHz Bandwidth (peak-to-peak value)		-	-	120	mV
Temperature Coefficient	0~50°C RH Non-Condensing		-	±0.03	-	%/°C
Stand-by Power Consump.	230VAC Input	12V Output	-	-	0.3	W
Short Circuit Protection			Hiccup, Continuous, Self-Recover			
Over-Current Protection			≥105%Io, Self-Recovery			
Over-Voltage Protection	12V Output		15-18V (Shutdown o/p Voltage)			
Minimum Load			0	-	-	%
Start-up Delay Time			-	-	0.5	s
Hold-up Time	115VAC		-	12	-	ms
	230VAC		-	30	-	
Connector			2-Way Screw Terminal			
Note: *The “parallel cable” method is used for ripple and noise test.						

Mechanical.

Case Material	Plastic, Heat-Resistant (UL94V-0)
Dimensions	90 × 35 × 58 mm (3.54 × 1.38 × 2.28 in) (H x W x D)
Weight	0.12 kg (0.26 lb)
Cooling	Free Air Convection

Specifications.

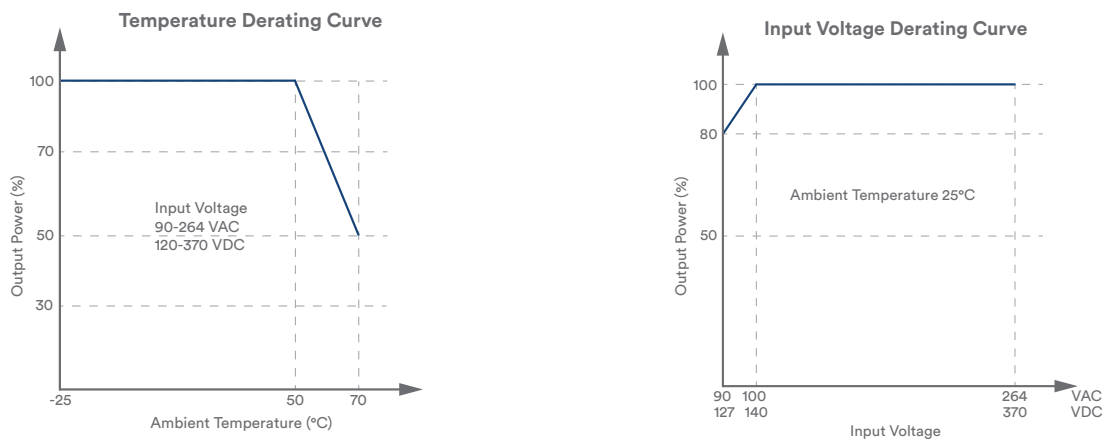
General.

Characteristics		Operating Conditions	Min.	Typ.	Max.	Unit
Isolation Test	Input-Output	Electric Strength Test for 1 min., (leakage current <5mA)	4000	-	-	VAC
Operating Temperature			-25	-	+70	°C
Storage Temperature			-40	-	+85	
Operating Humidity		Non-Condensing	10	-	95	%RH
Storage Humidity		Non-Condensing	10	-	95	%RH
Operating Altitude			-	-	5000	m
Switching Fequency			-	65	-	kHz
Power Derating	+50°C to +70°C		2.5	-	-	%/°C
	90VAC - 100VAC		2.0	-	-	%/VAC
Safety Standard			UL62368-1 / EN62368-1			
Safety Class			Class II			
MTBF		MIL-HDBK-217F @ 25°C	>900,000 hours			
Heat Dissipation			-	-	9	BTU/h

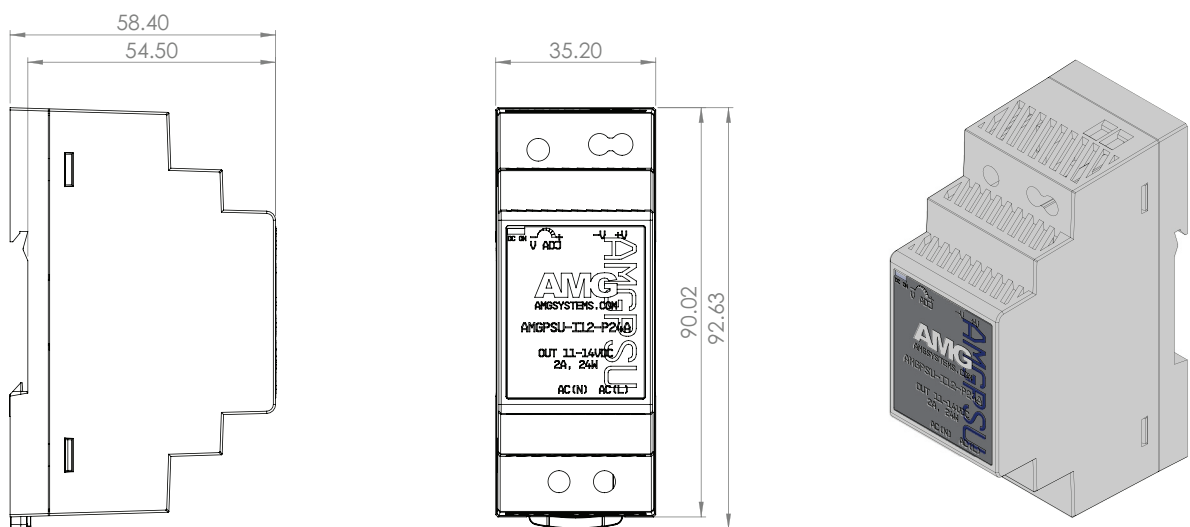
Regulatory.

Emissions	CE	CISPR32, BS EN / EN55032 Class B BS EN / EN 61000-3-2 BS EN / EN 61000-3-3
	RE	CISPR32, BS EN / EN55032 Class B
Immunity	ESD	BS EN / EN 61000-4-2 (Contact ±6KV / Air ±8KV)
	RS	BS EN / EN 61000-4-3 (10V/m)
	EFT	BS EN / EN 61000-4-4 (±2KV)
	Surge	BS EN / EN 61000-4-5 (Line to Line ±2KV)
	CS	BS EN / EN 61000-4-6 (10V r.m.s)
	PFMF	BS EN / EN 61000-4-8
	Voltage Dips, Short Interruptions and Voltage Variations Immunity	IEC/EN 61000-4-11 (0%, 70%)
Traffic		NEMA TS2
Supply Chain		NDAA Compliant

Product Characteristic Curve.



Product Dimensions.



Part Numbers.

24W Industrial DIN-Rail Power Supplies

AMGPSU-I12-P24A	Industrial DIN Rail Power Supply, 12V Nominal Output (10.8-13.8V Adjustable), 24W (2A)
AMGPSU-I12-P24A-K	Industrial DIN Rail Power Supply Kit, 12V, 24W (2A), DIN Rail, Mains & DC Cables Included*

* Region Specific Stripped Power Cord (UK, EU, US), 125mm (5in) DIN Rail, 400mm (16in) DC Power Cable (-K Models Only)

Unless otherwise specified, parameters in this datasheet were measured under the conditions of Ta=25°C. humidity <75% with nominal input voltage and rated output load.
In a continuing effort to improve and advance technology, product specifications are subject to change without notice. Please visit www.amgsystems.com for the latest product specifications.

